

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                                                                       |
|----------------------|---------------------------------------------------------------------------------------|
| Product Description: | <b>Diaminoethanetetra-acetic acid</b>                                                 |
| Cat No. :            | <b>Q23254</b>                                                                         |
| Synonyms             | (Ethylenedinitrilo)tetraacetic acid; EDTA; Edetic acid; Diaminoethanetetraacetic acid |
| Index No             | 607-429-00-8                                                                          |
| CAS No               | 60-00-4                                                                               |
| EC No                | 200-449-4                                                                             |
| Molecular Formula    | C10 H16 N2 O8                                                                         |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

|         |                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| Company | Thermo Fisher Scientific India Pvt. Ltd<br>403-404, B-wing, Delphi,<br>Hiranandani Business Park,<br>Powai, Mumbai 4000076, INDIA |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|

|                |                                     |
|----------------|-------------------------------------|
| E-mail address | laboratorysolution@thermofisher.com |
|----------------|-------------------------------------|

### 1.4. Emergency telephone number

India Toll Free: 18 00 22 22 30  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001 (202) 483-7616

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

Acute Inhalation Toxicity - Dusts and Mists  
Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (repeated exposure)

Category 4 (H332)  
Category 2 (H319)  
Category 2 (H373)

## **Environmental hazards**

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## **2.2. Label elements**



Signal Word

Warning

## **Hazard Statements**

H319 - Causes serious eye irritation

H332 - Harmful if inhaled

H373 - May cause damage to organs through prolonged or repeated exposure if inhaled

## **Precautionary Statements**

P280 - Wear eye protection/ face protection

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P312 - Call a POISON CENTER or doctor if you feel unwell

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P337 + P313 - If eye irritation persists: Get medical advice/attention

## **2.3. Other hazards**

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### **3.1. Substances**

| Component                               | CAS No  | EC No             | Weight % | CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567 |
|-----------------------------------------|---------|-------------------|----------|-----------------------------------------------------------------------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4 | EEC No. 200-449-4 | 99       | Eye Irrit. 2 (H319)<br>Acute Tox. 4 (H332)<br>STOT RE 2 (H373)                          |

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Obtain medical attention if irritation persists.        |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

QUAQ23254

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

## **6.2. Environmental precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

## **6.3. Methods and material for containment and cleaning up**

Sweep up and shovel into suitable containers for disposal.

## **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### **7.2. Conditions for safe storage, including any incompatibilities**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Technical Rules for Hazardous Substances (TRGS) 510**      Class 11  
**Storage Class (LGK) (Germany)**

### **7.3. Specific end use(s)**

Use in laboratories

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

### **8.1. Control parameters**

#### **Exposure limits**

List source(s):

#### **Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### **Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)**

See table for values; Workers

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

| Component                                                 | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-----------------------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Ethylenediamine tetraacetic acid (EDTA)<br>60-00-4 ( 99 ) | DNEL = 3mg/m <sup>3</sup>        |                                     | DNEL = 1.5mg/m <sup>3</sup>        |                                       |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                 | Fresh water    | Fresh water sediment | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|-----------------------------------------------------------|----------------|----------------------|--------------------|------------------------------------|--------------------------|
| Ethylenediamine tetraacetic acid (EDTA)<br>60-00-4 ( 99 ) | PNEC = 2.2mg/L |                      | PNEC = 1.2mg/L     | PNEC = 43mg/L                      | PNEC = 0.72mg/kg soil dw |

| Component                                                 | Marine water    | Marine water sediment | Marine water intermittent | Food chain | Air |
|-----------------------------------------------------------|-----------------|-----------------------|---------------------------|------------|-----|
| Ethylenediamine tetraacetic acid (EDTA)<br>60-00-4 ( 99 ) | PNEC = 0.22mg/L |                       |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Butyl rubber   | recommendations   |                 |             |                       |
| Nitrile rubber |                   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                                 |                                   |
|-----------------------------------------|---------------------------------|-----------------------------------|
| Physical State                          | Powder Solid                    |                                   |
| Appearance                              | White                           |                                   |
| Odor                                    | Odorless                        |                                   |
| Odor Threshold                          | No data available               |                                   |
| Melting Point/Range                     | 220 °C / 428 °F                 |                                   |
| Softening Point                         | No data available               |                                   |
| Boiling Point/Range                     | No information available        |                                   |
| Flammability (liquid)                   | Not applicable                  | Solid                             |
| Flammability (solid,gas)                | No information available        |                                   |
| Explosion Limits                        | No data available               |                                   |
| Flash Point                             | > 100 °C / > 212 °F             | Method - No information available |
| Autoignition Temperature                | 200 °C / 392 °F                 |                                   |
| Decomposition Temperature               | > 150°C                         |                                   |
| pH                                      | 2.5                             | 10 g/L (23°C)                     |
| Viscosity                               | Not applicable                  | Solid                             |
| Water Solubility                        | 0.5 g/L (20°C) - 2.2 g/L (80°C) |                                   |
| Solubility in other solvents            | No information available        |                                   |
| Partition Coefficient (n-octanol/water) |                                 |                                   |
| Vapor Pressure                          | 0.013 hPa @ 20 °C               |                                   |
| Density / Specific Gravity              | 0.86 @ 20°C                     |                                   |
| Bulk Density                            | No data available               |                                   |
| Vapor Density                           | Not applicable                  | Solid                             |
| Particle characteristics                | No data available               |                                   |

### 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | C10 H16 N2 O8          |
| Molecular Weight  | 292.23                 |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

## Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

## 10.4. Conditions to avoid

Avoid dust formation. Incompatible products. Excess heat.

## 10.5. Incompatible materials

Strong oxidizing agents. Strong bases. Metals. copper.

## 10.6. Hazardous decomposition products

Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Category 4

| Component                               | LD50 Oral                                 | LD50 Dermal | LC50 Inhalation |
|-----------------------------------------|-------------------------------------------|-------------|-----------------|
| Ethylenediamine tetraacetic acid (EDTA) | 4500 mg/kg ( Rat )<br>>2000 mg/kg ( Rat ) | -           | 1 mg/l (rat)    |

#### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

#### (c) serious eye damage/irritation;

Category 2

#### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

#### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

#### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

#### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

#### (h) STOT-single exposure;

Based on available data, the classification criteria are not met

#### (i) STOT-repeated exposure;

Category 2

Route of exposure

Inhalation

Target Organs

Respiratory system.

#### (j) aspiration hazard;

Not applicable

Solid

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

**Symptoms / effects, both acute and delayed** No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component                               | Freshwater Fish                                                                                                  | Water Flea                                   | Freshwater Algae                                 |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | LC50: 34 - 62 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 44.2 - 76.5 mg/L, 96h static (Pimephales promelas) | EC50: = 113 mg/L, 48h Static (Daphnia magna) | EC50: = 1.01 mg/L, 72h (Desmodesmus subspicatus) |

### 12.2. Persistence and degradability

#### Persistence

Soluble in water, Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.



# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

## Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### ADR

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### IATA

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

#### 14.5. Environmental hazards

No hazards identified

#### 14.6. Special precautions for user

No special precautions required.

#### 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                               | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------------------------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4 | 200-449-4 | -      | -   | X     | X    | KE-13648 | X    | X    |

| Component                               | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------------------------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

## Authorisation/Restrictions according to EU REACH

| Component                               | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                               | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | 60-00-4 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component                               | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------------------------------|---------------------------------------|-------------------------|
| Ethylenediamine tetraacetic acid (EDTA) | WGK2                                  |                         |

| Component                                                 | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Ethylenediamine tetraacetic acid (EDTA)<br>60-00-4 ( 99 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted

QUAQ23254

# SAFETY DATA SHEET

Diaminoethanetetra-acetic acid

Revision Date 11-Oct-2023

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H319 - Causes serious eye irritation

H332 - Harmful if inhaled

H373 - May cause damage to organs through prolonged or repeated exposure

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 15-Dec-2011

**Revision Date** 11-Oct-2023

**Revision Summary** Next Revision Date - Oct 2028.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**